Suzuki matrices and a chiral polytope

Complete treatment of the fixed F003 source

Human Compression project

This volume treats the four-page preprint *On the rank of Suzuki polytopes: an answer to Hubard and Leemans*, arXiv:2010.04080v1, 8 October 2020.

The first manuscript identifies the error in its matrix classification, proves the corrected matrix criterion, and retains its six algebraic cases and final generation argument with their exact hypotheses.

The second manuscript is the unchanged accepted construction of a chiral rank-four polytope of type $\{31, 31, 25\}$ with automorphism group Sz(32768). It refutes the source's principal theorem. Its exact finite-field certificate and prior mathematical reviews are retained by reference; the acquisition is not repeated.

The source conclusions are distinguished from explicitly attributed structural and reconstruction premises. This volume does not assign a compression factor, add a second refutation, or claim Lean verification. Completion concerns the specified source edition and does not assert a classification of all Suzuki chiral polytopes.

The machine-readable claim ledger and exact dependency list accompany the TeX sources. General proofs remain subject to distinct mathematical review until the review record is sealed.

Symmetric matrices in the Suzuki group and the scope of a rank argument

Human Compression project

Abstract

The proposed classification of symmetric Suzuki matrices in Spiga's four-page preprint imposes a product identity outside its stated domain. We identify the missing defect vector and give the correct matrix criterion. We also verify all six algebraic cases of the source: they classify the stronger system that was actually solved, rather than all symmetric Suzuki matrices. The final non-generation argument is valid with this additional hypothesis. Together with the previously accepted rank-four counterexample, reproduced unchanged in the accompanying volume, these results dispose of the mathematical claims in the specified source edition.

1 The source and the retained results

We examine Spiga's preprint *On the rank of Suzuki polytopes: an answer to Hubard and Leemans* [1], in its edition arXiv:2010.04080v1 of 8 October 2020. This is the four-page source registered as F003. Theorem 1.1 asserts that a chiral polytope with almost simple Suzuki automorphism group has rank three. The accepted construction [5] gives a chiral polytope of type $\{31, 31, 25\}$ with automorphism group Sz(32768) and therefore refutes that theorem. Its five-page proof, exact field certificate, and mathematical review are imported without alteration. No search or acquisition underlying that construction is repeated here.

The source has only one numbered theorem. Its remaining mathematical content consists of the reduction at the start of Section 3, a matrix calculation, six cases solving the resulting equations, and the concluding generation argument. The present note treats each of these. It does not assert that a later printing has the same wording.

We retain Wilson's characterization in precisely the form quoted on page 2 of the source: a matrix in $\mathrm{Sp}_4(q)$ belongs to $\mathrm{Sz}(q)$ if and only if it preserves the displayed semilinear product on *orthogonal pairs*. We also retain Suzuki's standard group structure: simplicity for $q \geq 8$, the single class of involutions, and the ovoid description in which each involution fixes exactly one of $q^2 + 1$ points, with $q - 1$ involutions fixing each point. These are attributed inputs from [4, 3], not conclusions of our finite checks. The reduction to the simple group and rank four in the first paragraph of source Section 3 is expressly attributed there to Hubard and Leemans [2, Theorem 1]. We retain that quoted implication as a premise; the full chapter is outside the accessible corpus (historical identifier L004), and is not credited as recovered here. Our counterexample already lies in the reduced case, so its refutation does not depend on this reduction.

2 The symmetric set and the defect vector

Put $q = 2^{2e+1}$ and $t = 2^e$, where $e \geq 1$, so that $2t^2 = q$. This agrees with the calculation on source page 4. On page 1 the additional equality $t = \sqrt{2q}$ and the description of $\mathbb{F}_q^4$ as a vector space over $\mathbb{F}_4$ are typographical errors; the consistent values are $t = \sqrt{q/2}$ and base field $\mathbb{F}_q$. Let J be the antidiagonal matrix with all four antidiagonal entries equal to one, and put

$f(u,v) = u^{\mathsf{T}} J v$ on $V = \mathbb{F}_q^4$. The product in the source is alternating, additive in both variables, and satisfies

$$(\lambda u) \bullet (\mu v) = \lambda^t \mu^t (u \bullet v).$$

Its nonzero values on unordered distinct basis pairs are

$$e_1 \bullet e_2 = e_2, \quad e_1 \bullet e_3 = e_4, \quad e_2 \bullet e_4 = e_1, \quad e_3 \bullet e_4 = e_3. \tag{1}$$

Both remaining distinct basis products are zero. In particular, $f(e_1, e_4) = f(e_2, e_3) = 1$, although both products are zero.

Lemma 1. *Let K be a group and let $j \in K$ satisfy $j^2 = 1$. The map $x \mapsto xj$ is a bijection*

$$\{x \in K : xjx = j\} \longrightarrow \{h \in K : h^2 = 1\}.$$

If $K \leq \mathrm{Sp}_4(q)$ and $j = J$, its domain consists exactly of the symmetric matrices in K.

Proof. The equation $xjx = j$ is equivalent to $(xj)^2 = 1$, and the inverse map is $h \mapsto hj$. For a symplectic matrix, $x^{\mathsf{T}} J x = J$. Comparing this with $xJx = J$, and cancelling the invertible matrix Jx, gives $x^{\mathsf{T}} = x$. Conversely a symmetric symplectic matrix satisfies $xJx = J$. □

Thus, writing $X = \{x \in \mathrm{Sz}(q) : xJx = J\}$, the retained involution counts give

$$|X| = 1 + (q^2 + 1)(q - 1) = q^3 - q^2 + q. \tag{2}$$

The source's proposed list has only q elements. This elementary discrepancy already occurs in the accepted counterexample paper; it diagnoses the matrix classification independently of the polytope intersection and chirality conditions.

Lemma 2. *Suppose that $D : V \times V \to V$ is alternating and additive in each variable, with $D(\lambda u, \mu v) = \lambda^t \mu^t D(u, v)$. It vanishes on all orthogonal pairs if and only if there is a vector $r \in V$ such that*

$$D(u, v) = f(u, v)^t r \qquad (u, v \in V). \tag{3}$$

Proof. The orthogonal basis pairs force $D_{12} = D_{13} = D_{24} = D_{34} = 0$, where $D_{ij} = D(e_i, e_j)$. The pair $(e_1 + e_2, e_3 + e_4)$ is orthogonal and gives $D_{14} = D_{23}$. Denote this common value by r. Expanding in the basis yields

$$D(u, v) = (u_1 v_4 + u_4 v_1 + u_2 v_3 + u_3 v_2)^t r,$$

which is (3). Conversely (3) vanishes whenever $f(u, v) = 0$. □

For $g \in \mathrm{Sp}_4(q)$ take $D_g(u, v) = gu \bullet gv - g(u \bullet v)$. It has the hypotheses of Lemma 2. Consequently Wilson's condition says that the four orthogonal basis defects vanish and the two other defects are equal. It does not say that the latter defects vanish separately.

3 The equations that are valid

For ease of comparison set

$$x = \begin{pmatrix} A & B & C & D \\ B & E & F & G \\ C & F & H & I \\ D & G & I & K \end{pmatrix}. \tag{4}$$

The letter K here denotes an entry, not a group. The four orthogonal basis pairs in (1) give exactly the following ten distinct scalar equations. The numbers are the source numbers.

$$(BG + DE)^t = B, \tag{3s}$$
$$(AE + B^2)^t = E, \tag{4s}$$
$$(CG + DF)^t = F, \tag{5s}$$
$$(AF + BC)^t = G, \tag{6s}$$
$$(BI + DF)^t = D, \tag{7s}$$
$$(CI + DH)^t = I, \tag{8s}$$
$$(AH + C^2)^t = K, \tag{9s}$$
$$(EK + G^2)^t = A, \tag{10s}$$
$$(FK + GI)^t = C, \tag{11s}$$
$$(HK + I^2)^t = H. \tag{12s}$$

For example, the four coordinates of $xe_1 \bullet xe_2 = xe_2$ give (3s)–(6s). The pair (e_1, e_3) gives (7s), (6s), (8s), (9s); the pair (e_2, e_4) gives (10s), (3s), (11s), (7s); and (e_3, e_4) gives (11s), (5s), (12s), (8s). This also specifies the omitted computations on source page 3.

The two nonorthogonal defects, after applying the inverse of the Frobenius map $z \mapsto z^t$ coordinatewise, are

$$\begin{pmatrix} BK + DG \\ AG + BD \\ CK + DI \\ AI + CD \end{pmatrix}, \qquad \begin{pmatrix} EI + FG \\ BF + CE \\ FI + GH \\ BH + CF \end{pmatrix}. \tag{5}$$

Their equality consists of four off-diagonal equations of $xJx = J$. For instance the second equality is the $(1, 2)$ entry, and the other three are the $(2, 4)$, $(3, 4)$ and $(1, 3)$ entries. Thus a symmetric symplectic matrix already has equal defects.

Proposition 3. *A symmetric matrix x belongs to* $\mathrm{Sz}(q)$ *if and only if $xJx = J$ and (3s)–(12s) hold.*

Proof. Necessity follows from the form and the four orthogonal basis products. Conversely $xJx = J$ implies invertibility and, by symmetry, membership in $\mathrm{Sp}_4(q)$. Equations (3s)–(12s) make the four orthogonal basis defects zero. Equation (5) and $xJx = J$ make the remaining two equal. Lemma 2 and Wilson's retained characterization now give $x \in \mathrm{Sz}(q)$. □

Source equations (13)–(20) instead require both vectors in (5) to be zero. That is a stronger condition and is the substantive error. The repeated $xe_1 \bullet xe_3$ immediately before $x(e_2 \bullet e_4)$ on page 3 is a separate index error: the intended pair is (e_2, e_4). Equations (21)–(22) are valid:

$$D^2 + CG + BI + AK = 1, \qquad CG + F^2 + EH + BI = 1. \tag{6}$$

They are the $(1, 4)$ and $(2, 3)$ entries of $xJx = J$.

Proposition 4. *For every allowed q, the matrix*

$$x_0 = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \tag{7}$$

belongs to X but violates source equation (14).

Proof. Direct multiplication gives $x_0^{\mathsf{T}} = x_0$ and $x_0 J x_0 = J$. The four orthogonal basis defects are zero; both other defects equal e_2. These are calculations over $\mathbb{F}_2$, so the exponent t changes none of their entries. Lemma 2 proves preservation on every orthogonal pair, hence membership in the source Suzuki model. But source equation (14), $BD = AG$, reads $1 = 0$ for this matrix. This refutes the intermediate classification uniformly; it is not itself a polytope marking. □

4 What the six source cases prove

Let Z be the set of symmetric matrices (4) satisfying (3s)–(12s), (6), and the eight additional equations

$$DG = BK, \quad BD = AG, \quad CK = DI, \quad AI = CD,$$
$$EI = GF, \quad BF = EC, \quad FI = GH, \quad BH = CF.$$

These are source equations (13)–(20), in order. Define

$$T = \{\mathrm{diag}(a, a^{2t+1}, a^{-2t-1}, a^{-1}) : a \in \mathbb{F}_q^*\}. \tag{8}$$

Theorem 5. *The solutions of the stronger system are exactly $Z = \{J\} \cup T$. Thus all six source cases are valid as a classification of Z. This set is a proper subset of X.*

Proof. We retain the order of the cases and record the equations used.

Suppose first that $G \neq 0$. Equations (13), (14), (17), (19) give

$$D = BK/G, \quad A = B^2K/G^2, \quad F = EI/G, \quad H = EI^2/G^2. \tag{9}$$

In the first equation of (6), the terms D^2 and AK cancel, leaving $CG + BI = 1$. Equation (15) gives $CK = BIK/G$. Multiplication of $CG + BI = 1$ by K/G therefore gives $K/G = 0$, so $K = D = A = 0$. Similarly (18) and (9) give $EC = BEI/G$; multiplication by E/G gives $E/G = 0$. Equation (4s) now gives $B = 0$, and (9) gives $H = 0$. Equation (9s), with $K = 0$, gives $C = 0$. This contradicts $CG + BI = 1$.

Suppose $G = 0$ and $B \neq 0$. Equations (13)–(14) give $K = D = 0$. Equation (3s) then gives $B = 0$, a contradiction. This is source Case 3.2; its intervening consequence $BI = 1$ from (6) is correct but is not needed for the contradiction.

Suppose $G = B = 0$ and $I \neq 0$. Equation (17) gives $E = 0$. The second equation of (6) gives $F^2 = 1$, whence $F = 1$ in characteristic two. Equation (19) gives $FI = 0$, a contradiction.

Suppose $G = B = I = 0$ and $C \neq 0$. Equation (15) gives $K = 0$. The first equation of (6) gives $D = 1$. Equation (16) then gives $CD = 0$, a contradiction.

Suppose $G = B = I = C = 0$ and $D \neq 0$. Equation (3s) gives $E = 0$; the second equation of (6) gives $F = 1$. Equation (6s) gives $A = 0$, and the first equation of (6) gives $D = 1$. Equation (8s) gives $H = 0$, and (9s) gives $K = 0$. The resulting matrix is J.

Finally suppose $G = B = I = C = D = 0$. The first equation of (6) gives $AK = 1$, so $A \neq 0$ and $K = A^{-1}$. Equation (5s) gives $F = 0$; the second equation of (6) gives $EH = 1$. Equation (10s) becomes $E^t = A^{t+1}$. Raising to the power $2t$, using $2t^2 = q$ and $z^q = z$ in $\mathbb{F}_q$, gives $E = A^{2t+1}$. Thus $H = A^{-2t-1}$, and the matrix belongs to T. These six cases exhaust all choices of the five entries tested successively.

For the converse, J satisfies every displayed equation by substitution. A diagonal member of T has $AK = EH = 1$. The only nonzero product equations are (4s), (9s), (10s), (12s). For example

$$(AE)^t = a^{2t^2+2t} = a^{q+2t} = a^{2t+1} = E, \qquad (AH)^t = a^{-2t^2} = a^{-1} = K.$$

The other two follow by inversion. Hence each listed matrix lies in Z. Each is also symplectic, so Proposition 3 shows $Z \subseteq X$. Proposition 4 shows that the inclusion is proper. □

5 The generation argument with its exact hypothesis

We first retain the valid normalization preceding the source calculation. Suppose $\sigma_1\sigma_2$, $\sigma_2\sigma_3$, and $\sigma_1\sigma_2\sigma_3$ are involutions in $\mathrm{Sz}(q)$. Conjugacy of involutions allows the last product to be J. Then $\sigma_1\sigma_2 = J\sigma_3^{-1}$ and $\sigma_2\sigma_3 = \sigma_1^{-1}J$. Squaring these products gives

$$\sigma_1^{-1}, \sigma_3^{-1} \in X. \tag{10}$$

Every step through (10) is valid. Replacing X by Z is not.

Proposition 6. *Under these rotation relations and this normalization, if* $\sigma_1^{-1}, \sigma_3^{-1} \in Z$*, the three rotations do not generate* $\mathrm{Sz}(q)$*.*

Proof. If $\sigma_1^{-1} = J$, then $\sigma_2\sigma_3 = 1$; if $\sigma_3^{-1} = J$, then $\sigma_1\sigma_2 = 1$. Both contradict the requirement that the adjacent products have order exactly two. Theorem 5 therefore places both endpoint rotations in T. This is an abelian subgroup of order $q-1$, and direct conjugation by J inverts every element. Consequently

$$\langle \sigma_1, \sigma_2, \sigma_3 \rangle = \langle \sigma_1, \sigma_3, J \rangle \leq T \rtimes \langle J \rangle.$$

The containing group has order $2(q-1)$ and is soluble: its commutator subgroup is contained in T. It cannot equal the nonabelian simple group $\mathrm{Sz}(q)$. □

This recovers source Section 3.7 as a conditional statement. The accepted rank-four marking satisfies the genuine relations and the polytope intersection property; after normalization its endpoint matrix fails equation (13) of the stronger system. The missing hypothesis is therefore a genuine restriction on the construction, not an alternative coordinate convention.

6 Attribution and disposition

The equations from orthogonal pairs, the six successive algebraic cases, and the torus non-generation argument originate in the source. We have retained them with the correct hypotheses and supplied the omitted coordinate checks and the converse to the six-case solution. The involution-translate observation and defect-vector argument are reused from the project's earlier F003 audit; the accepted marking and its diagonal subfield intersection proof are reused unchanged from [5]. Proposition 3 assembles these observations into the corrected matrix criterion.

Wilson's characterization, Suzuki group structure, the quoted Hubard–Leemans reduction, and the polytope reconstruction theorem used in the counterexample remain explicit attributed premises. The source's principal rank theorem and intermediate classification are false; their disposition is by counterexample, not by recovery of their conclusions. No classification of every chiral Suzuki polytope, no formal Lean proof, and no compression factor is claimed. The binary calculation accompanying (7) checks only that small matrix identity; general proofs in this note require mathematical review. This supplement adds no principal-refutation count.

References

[1] P. Spiga, *On the rank of Suzuki polytopes: an answer to Hubard and Leemans*, arXiv:2010.04080v1, 8 October 2020.

[2] I. Hubard and D. Leemans, *Chiral polytopes and Suzuki simple groups*, Rigidity and Symmetry, Fields Institute Communications 70 (2014), 155–175.

[3] M. Suzuki, *On a class of doubly transitive groups*, Annals of Mathematics 75 (1962), 105–145.

[4] R. A. Wilson, *A new approach to the Suzuki groups*, Mathematical Proceedings of the Cambridge Philosophical Society 148 (2010), 425–428.

[5] Human Compression project, *A chiral* 4*-polytope with automorphism group* Sz(32768), accepted F003 correction, public proof edition v1, `corrections/suzuki-rank/v1/proof`.

A chiral 4-polytope with automorphism group Sz(32768)

Human Compression project

Abstract

We give explicit matrices for a chiral polytope of type $\{31, 31, 25\}$ whose automorphism group is $\mathrm{Sz}(2^{15})$. Its facet and vertex-figure groups are two conjugate subfield groups $\mathrm{Sz}(2^5)$, meeting in a split torus of order 31. The construction contradicts Theorem 1.1 in the specified preprint edition of Spiga's paper on Suzuki polytopes. All finite-field calculations are supplied in a short executable certificate. The proof of generation and of the intersection property uses the classical structure of Suzuki groups, not an enumeration of the ambient group.

Spiga's preprint [1], arXiv:2010.04080v1 of 8 October 2020, asserts that every chiral polytope whose automorphism group lies between a Suzuki simple group and its automorphism group has rank three. The question is attributed there to Hubard and Leemans [2]. The theorem below concerns that precise source edition. It does not assert that every later printing has the same text, or make a separate priority claim about the subsequent literature.

Theorem 1. *There exists a chiral abstract polytope of rank four and type* $\{31, 31, 25\}$ *with automorphism group* $\mathrm{Sz}(32768)$.

1 Coordinates and retained group structure

Let $k = \mathbb{F}_{2^{15}}$ and $\theta = 2^8$. We use the standard Suzuki matrices

$$U(x,y) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ x & 1 & 0 & 0 \\ x^{\theta+1} + y & x^{\theta} & 1 & 0 \\ x^{\theta+2} + xy + y^{\theta} & y & x & 1 \end{pmatrix}, \qquad H(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}),$$

where $x, y \in k$ and $t \in k^*$. Let J be the matrix with ones on the antidiagonal and zeroes elsewhere. These matrices generate $G = \mathrm{Sz}(k)$; see, for example, the coordinate description in Bray and Bäärnhielm [4]. Write $K = \mathbb{F}_{32}$ and $F = \mathrm{Sz}(K)$ for the standard subfield group. The restriction of the map $x \mapsto x^{\theta}$ to K is $x \mapsto x^8$, so the same formulas define F.

We use the following classical facts from Suzuki's subgroup classification [3]. The group $\mathrm{Sz}(q)$ is simple for $q \geq 8$, has order $q^2(q^2+1)(q-1)$, and acts doubly transitively on its ovoid. A nonidentity element of a split torus fixes exactly the two points defining that torus. The maximal subgroups are point stabilizers, the three types of odd-torus normalizers, and maximal Suzuki subfield groups. Their relevant orders are

$$q^2(q-1), \qquad 2(q-1), \qquad 4(q+\sqrt{2q}+1), \qquad 4(q-\sqrt{2q}+1).$$

A subfield group $\mathrm{Sz}(2^d)$ is maximal when the odd quotient of field exponents is prime. The centralizer of a nonidentity odd-order element is cyclic of odd order. Finally, $\mathrm{Out}(\mathrm{Sz}(2^N)) \cong C_N$ for odd N. These are borrowed structural inputs. None is inferred from the finite computation below.

The standard Bruhat decomposition, also retained, is

$$\mathrm{Sz}(K) = U(K)H(K) \;\sqcup\; U(K)H(K)JU(K). \tag{1}$$

The first cell is lower triangular. In the second cell the entry in row one, column four is nonzero. Both assertions follow immediately by multiplying the displayed matrices.

2 The explicit rotations

Take z to be the residue class of X in

$$k = \mathbb{F}_2[X]/(X^{15} + X + 1), \qquad \eta = z^{1057}. \tag{2}$$

The polynomial is irreducible, z has order 32767, and η has order 31. Thus $K = \mathbb{F}_2(\eta)$. Set

$$\lambda = z^{11} + z^7 + z^6 + z^4 + z^3 + z + 1. \tag{3}$$

For reference, its minimal polynomial over K is

$$T^3 + \eta^{12}T^2 + \eta^{11}T + \eta. \tag{4}$$

In particular $\lambda \notin K$. Define the involution

$$i(x, y) = U(x, y)^{-1} J U(0, 1) J U(x, y)$$

and put

$$\begin{aligned} &u = i(\eta^2, \eta^3), \quad v_0 = i(\eta^5, \eta^{30}), \\ &h = H(\lambda), \qquad v = h v_0 h^{-1}, \qquad b = H(\eta), \\ &a = ub^{-1}, \qquad\quad c = b^{-1}v. \end{aligned} \tag{5}$$

These formulas, together with (2)–(3), specify every entry without a choice of a primitive element or of an unspecified root.

Lemma 2. *The elements in (5) satisfy*

$$|a| = 31, \quad |b| = 31, \quad |c| = 25, \qquad |ab| = |bc| = |abc| = 2. \tag{6}$$

Neither u nor v_0 is upper triangular, lower triangular, or antidiagonal.

Proof. The assertions are exact calculations in the field (2). We give the calculation in executable form as `check_arithmetic.py`; it uses only polynomial arithmetic over $\mathbb{F}_2$ and multiplication of 4×4 matrices. An integer $m = \sum m_j 2^j$ denotes $\sum m_j z^j$. In this notation $\eta = 624$ and $\lambda = 2267$, and the two involutions occurring before conjugation are

$$u = \begin{pmatrix} 4671 & 4175 & 5401 & 1 \\ 5156 & 1562 & 1830 & 5401 \\ 845 & 5157 & 1562 & 4175 \\ 1131 & 845 & 5156 & 4671 \end{pmatrix}, \quad v_0 = \begin{pmatrix} 5401 & 5401 & 4867 & 1 \\ 4467 & 1563 & 1562 & 4867 \\ 1131 & 5717 & 1563 & 5401 \\ 624 & 1131 & 4467 & 5401 \end{pmatrix}. \tag{7}$$

Reduction modulo $X^{15} + X + 1$ gives $a^{31} = b^{31} = c^{25} = I$, with $a, b, c^5 \neq I$. It also gives $u^2 = v^2 = (abc)^2 = I$, with all three matrices nonidentity. The identities $ab = u$ and $bc = v$ follow from the definitions. The off-diagonal and diagonal entries in (7) establish the last assertion directly.

For reproducibility, the certificate first checks irreducibility by $z^{2^{15}} = z$ and the two polynomial gcd tests for the prime divisors $3, 5$ of 15. It verifies primitivity by the prime divisors $7, 31, 151$ of 32767, reconstructs every matrix from (2)–(5), and checks all the equalities just stated. It also verifies the Suzuki and symplectic matrix equations and rejects a deliberately altered entry of v. The earlier search output is not read by this checker. □

3 The two face groups and their intersection

Lemma 3. *We have* $\langle a, b\rangle = F$ *and* $\langle b, c\rangle = hFh^{-1}$.

Proof. The first group equals $\langle u, b\rangle \leq F$. At $q = 32$, the two nonsplit torus normalizers have orders 100 and 164, and the proper Suzuki subfield group is Sz(2), of order 20. None has order divisible by 31. A proper subgroup containing b must therefore lie in a point stabilizer or in the normalizer of a split torus. The only ovoid points fixed by b are the standard points $0, \infty$. Their stabilizers are upper and lower triangular, respectively. The normalizer of $\langle b\rangle = H(K)$ consists of the diagonal matrices and their antidiagonal coset. Lemma 2 excludes every possibility for a proper subgroup containing u, b. This proves the first assertion.

The element h centralizes b. Consequently $h^{-1}\langle b, c\rangle h = \langle b, v_0\rangle$. The same argument, with v_0 in place of u, proves the second assertion. □

Lemma 4. *For the diagonal conjugator in (5),*

$$F \cap hFh^{-1} = H(K) = \langle b\rangle. \tag{8}$$

Proof. The diagonal subgroup $H(K)$ is contained in the intersection. Suppose $g \in F \cap hFh^{-1}$. Then both g and $h^{-1}gh$ have entries in K. If g belongs to the second Bruhat cell in (1), its $(1, 4)$ entry is in K^*, whereas conjugation by h multiplies this entry by $\lambda^{-2(\theta+1)}$. Hence $\lambda^{514} \in K$. Since $\gcd(514, 32767) = 1$, this implies $\lambda \in K$, a contradiction.

Thus $g = U(x, y)H(t)$ with $x, y \in K$ and $t \in K^*$. Direct conjugation gives

$$h^{-1}U(x, y)h = U(x\lambda^{\theta}, y\lambda^{\theta+2}). \tag{9}$$

The root parameters in (1) are unique, so the two parameters on the right belong to K. If $x \neq 0$, then $\lambda^{256} \in K$; if $x = 0$ and $y \neq 0$, then $\lambda^{258} \in K$. Both are impossible because $\gcd(256, 32767) = \gcd(258, 32767) = 1$. Therefore $x = y = 0$ and $g \in H(K)$, as required. □

Lemma 5. *The three rotations generate* $G = \mathrm{Sz}(32768)$.

Proof. They generate both face groups from Lemma 3. The first, $F = \mathrm{Sz}(32)$, is a maximal subgroup of G because the quotient of field exponents is $15/5 = 3$. Equation (8) implies that the second face group is distinct from F. Hence the generated subgroup properly contains F and must equal G. □

4 The polytope and its chirality

We recall the rank-four rotation criterion of Schulte and Weiss [5], in the explicit form recorded in Leemans, Moerenhout and O'Reilly-Regueiro [6, §2, (C3′)]. Given generators a, b, c of finite orders at least two, with ab, bc, abc involutions, the required intersection condition is

$$\langle a\rangle \cap \langle b\rangle = 1, \qquad \langle b\rangle \cap \langle c\rangle = 1, \qquad \langle a, b\rangle \cap \langle b, c\rangle = \langle b\rangle. \tag{10}$$

These conditions give a chiral or directly regular abstract polytope. It is chiral precisely when there is no involutory automorphism α of the generated group satisfying

$$\alpha(a) = a^{-1}, \qquad \alpha(b) = a^2b, \qquad \alpha(c) = c. \tag{11}$$

This is a borrowed reconstruction theorem; the group and marking constructed in (5) are the new input.

Proof of Theorem 1. The rotation relations and generation follow from Lemmas 2 and 5. The groups $\langle a\rangle$ and $\langle b\rangle$ have prime order 31. If they had nontrivial intersection, they would coincide, and their product ab would have odd order, contrary to (6). Thus the first equality in (10) holds. The second follows from $\gcd(31, 25) = 1$, and the third is Lemma 4.

It remains to exclude (11). The outer automorphism group of G has odd order 15, so every involutory automorphism is inner. Write $\alpha(g) = s^{-1}gs$. Since G is centreless, $\alpha^2 = 1$ implies $s^2 = 1$. The automorphism cannot be the identity, because $|a| = 31$, so s would be an involution. But $\alpha(c) = c$ gives $s \in C_G(c)$, and this centralizer has odd order because $|c| = 25$. This is impossible. The polytope is therefore chiral, with full automorphism group G and type $\{31, 31, 25\}$. □

The computation also checks the two small dihedral groups $\langle a, bc\rangle$ and $\langle ab, c\rangle$, of orders 62 and 50, and their intersections with the face groups. Their mutual intersection has order two and is generated by abc. These finite checks corroborate (10); they do not replace the proof of (8) or the maximal-subgroup argument. The ambient group, of order 37 777 778 976 635 853 209 600, is never enumerated.

5 Comparison with the source coordinates

For a direct comparison with [1], let P be the permutation matrix for (1 2)(3 4). It preserves J and changes our torus to

$$PH(t)P^{-1} = \operatorname{diag}(t, t^{\theta+1}, t^{-\theta-1}, t^{-1}).$$

Our Suzuki graph map takes the minors with row and column pair order $(12, 13, 24, 34)$, then raises entries to 2^7. Conjugating by P changes that order to $(24, 12, 34, 13)$, exactly the order in Wilson's product as displayed on page one of [1]. This verifies the change of model rather than assuming identical coordinate conventions.

The certificate constructs $g \in G$ with $g^{-1}(abc)g = J$. In the source basis, the inverse of the first normalized rotation is

$$x = P(g^{-1}ag)^{-1}P^{-1} = \begin{pmatrix} 29124 & 17285 & 29707 & 20757 \\ 17285 & 3002 & 31396 & 7939 \\ 29707 & 31396 & 8273 & 19487 \\ 20757 & 7939 & 19487 & 23604 \end{pmatrix}. \tag{12}$$

The integer encoding is the one used in (7). This matrix is symmetric, belongs to the Suzuki group in the source model, and satisfies $xJx = J$. Nevertheless its entries give

$$x_{14}x_{24} + x_{12}x_{44} = 28331 \neq 0. \tag{13}$$

Thus it violates equation (13) of [1], whose right-hand side would be zero.

The reason is visible before that equation is imposed. The source states Wilson's condition $gx \bullet gy = g(x \bullet y)$ only for pairs with symplectic pairing zero. On page three it also imposes this condition on (e_1, e_4) and (e_2, e_3), both of which have pairing one. For the concrete matrix (12),

$$(xe_1) \bullet (xe_4) = (3245, 26415, 3338, 22277)^{\mathsf{T}} \neq 0, \qquad e_1 \bullet e_4 = 0.$$

The admissible orthogonal-pair identities do hold. The failure is therefore an application outside the stated domain of the product identity, not a discrepancy between the two matrix models.

There is also a count that detects the same defective classification. For $\mathcal{X} = \{x \in G : xJx = J\}$, the map $x \mapsto xJ$ is a bijection from $\mathcal{X}$ to the involutions together with the identity. Each of the q^2+1 ovoid points is fixed by $q-1$ involutions, and each involution has a unique fixed point. Hence $|\mathcal{X}| = (q^2 + 1)(q - 1) + 1$, whereas the classification in the source would give $|\mathcal{X}| = q$.

This observation diagnoses the source argument; Theorem 1 additionally verifies the polytope intersection and chirality conditions.

The contribution of this note is the explicit marking, the diagonal subfield intersection argument as applied to it, and the resulting counterexample to the stated rank bound. The matrix model, Suzuki subgroup classification and polytope reconstruction theorem are retained inputs with the attributions above. No whole-source acceptance, independent review, Lean verification, or compression measurement is claimed by this author packet itself.

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